Arizona Solar ~ Limp Grid Energy:

A Solar Reality Check for Arizona Utility Consumers

Mark Ketring WDD 130 A1 ~ Web Fundamentals ~ 4-July-2025

Introduction

My final project for WDD-130 is an informational website titled *Limp Grid Energy*, which explores a “cross-contamination” of solar energy, regulatory policy and my professional observations. The site is designed to educate, provoke thought, and share my journey as both a certified solar energy professional and an early-adopter, private solar system owner in Arizona.

The purpose of my website is twofold: First, to introduce some essential history and review my ongoing observations and background in renewable energy systems integration.

Second, to expose the complex and often frustrating regulatory environment that governs solar adoption in Arizona. The site includes a home page with a detailed solar policy timeline that outlines the evolution of Arizona’s solar regulations from 2006 to the present. The home page also has an online calculator that demonstrates the billing and offset disparities between the three largest utility providers in Arizona APS, TEP and SRP.
SRP being the largest unregulated producer-provider in Arizona.

The intended audience for this site includes, but is not limited to homeowners considering solar, energy policy advocates, employee associates, and professionals in the renewable energy industry. It also serves as a platform for those who may be unaware of how utility-driven politics and regulatory history can diminish the promises and deployments of renewable energy for end-users. The tone is frank, critically informative and candid, reflecting my strong personal views and reviews regarding background observations on the subject matter.

My final website for submission is hosted on GitHub Pages and can be accessed at:
https://mketbyui.github.io/wdd130/solarutility/index-solar_.html

Note: My original site address ended with /index-solar.html but that remains detached from it’s CSS association so it could be graded on the merit of the HTML requirements during week 10.

Ultimately I will lose the “_underscore” after solar and revert to the original url.

Goals and Objectives

My outcome goal for Arizona Solar ~ Limp Grid Energy was to create a site that is historically accurate, technically clear, emotionally engaging and intellectually stimulating. I wanted to:

- Share my personal and professional outlook on solar energy in AZ
- Educate potential adopters on Arizona's solar regulatory challenges
- Relate critique and satirical explanations and highlight the challenges of navigating utility rate plans and the certainty of future rate plan evolution.
- Develop basic web development skills using HTML and CSS
- Build the foundation for a user friendly annual offset calculator and an overview timeline component for use in future commercial projects that can be adapted to other states and their corresponding jurisdictions.

Ultimately, I set out to create a site that not only fulfilled the course requirements but now serves as a meaningful draft that I will leverage as a future resource for myself and others, as it documents practical examples and viewpoints on solar energy challenges that are also relevant across the country, not just "Arizona". Most sales consultants avoid frank and honest information exchange as they normally hold back or divert consumer questions and attention away from conflicted issues that might scuttle a sale. This is an ongoing problem that I have continuously engaged for 25 years in the energy management and system fulfillment fields.

Design Processing

My mental design processes began with a very clear vision: I wanted the site to feel personal yet professional, with a clean layout that supports long-form content. I started with very simple wireframes for each page, which were flexible, but with limited visuals that usually dominate renewable energy websites with fluff and stuff that appeals to feel good emotions. I intentionally chose a bold color palette that relates to the colors of Arizona's state flag. Keeping focus on the contextual history, by using relevant visuals.

My "B-LOG" page was actually an afterthought that hosts excess background information that will be too tedious for some consumer level layperson readers to wade through, but is there and available for detail minded readers to understand as strong opinion based supplemental material.

The solar policy timeline was the most content-relevant section. I used semantic HTML headings and paragraphs, with bold comments and some dry-type humor to clarify the regulatory history. I also included a call-to-action linking to a simple solar ROI calculator, reinforcing the site's practical value for any level of personal interaction capabilities.

One major design decision was to embrace a narrative tone—mixing technical facts with my own personal opinion and critical commentary. This gives the site my own unique voice and helped me differentiate it from generic sales or info-matic / info-magic / lead-generation solar pages.

Conclusion

This project has been a very relevant first-dive into both web development and digital story development. I've learned how to structure and style a very simple multi-page site, how to troubleshoot layout and path issues, and how to manage CSS toward creating visually appealing presentation. More or most importantly, I've learned that I can personally/professionally leverage web development as a super platform for advocacy and education. Completing this project was extremely challenging but rewarding. First-hand experience in dealing with GitHub was not fun as I figured out that it is very finicky and structured, to which I needed to conform to the platform, not the other way around. I also got a good handle on validating code. Which was faster but mentally reminiscent of pushing a stack of Fortran punch cards into a reader while praying quietly for a successful pass.

I am keenly aware that these skills need to be advanced in my own wheelhouse and will be most valuable if I continue to explore web development and integrate it into my work with renewable energy. Whether I'm building internal tools for my team, creating educational content for consumers, or launching a new entrepreneurial endeavor, I now have the foundation to move with confidence and clarity toward better understanding and utilization. Last of all I would like to say that this is a firm foundation for future impact in an industry that is starving for reconciliation with a disillusioned public sentiment that holds solar sales processing in the same regard as used car sales processing and lemon-law outcomes that don't meet pre-conceived expectations.

For this part of the project, you are required to type a final report using Word or PDF, detailing the design and development of your completed website. Your report must be at least 500 words.

Introduction (12 pts - 3 pts for each required subtopic)

- Separate Title page with your Title, your name and class, and the date.
- Introduce the subject of your website.
- Explain who the audience of your website is.
- Include the URL for your completed website.

Goals/Objectives (3 pts)

- Explain what you hope to accomplish with this website.

Design Process (5 pts)

- Describe how you came up with the design for your website or your process for coming up with your design or any major changes you made in your site?

Conclusion (5 pts)

- Describe what you learned from completing this project and contemplate how these skills may help you in the future.